

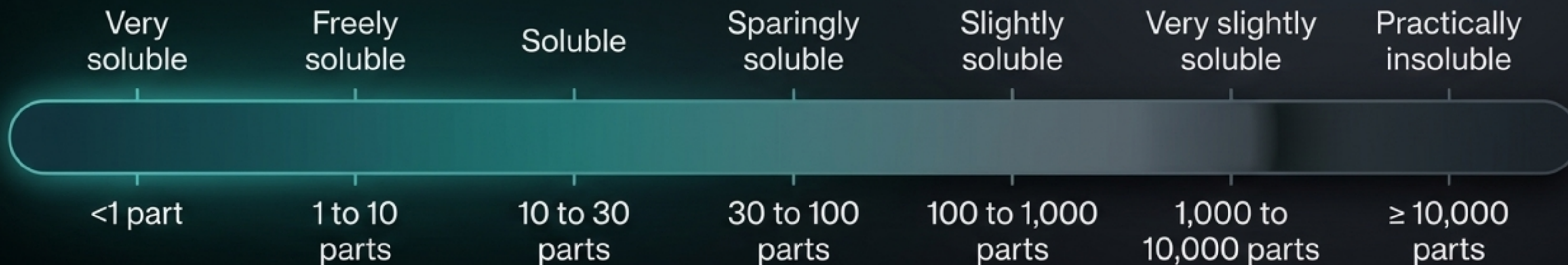
SOLUBILITY & DISTRIBUTION PHENOMENA

The physical chemistry of solvent-solute interactions

Based on the work of Dr. Ayat Allam

Defining the Spectrum of Solubility

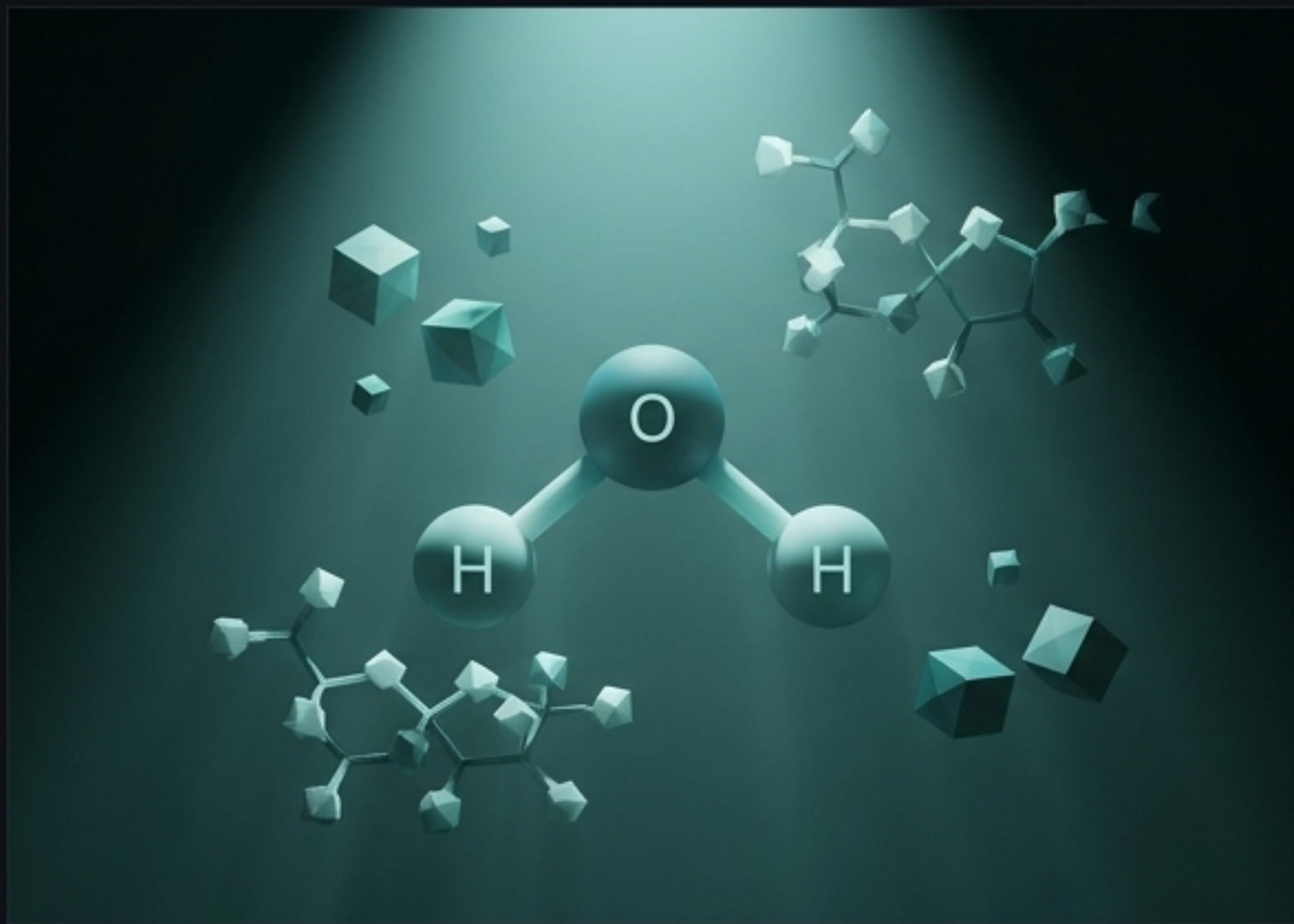
Solubility is the spontaneous interaction of substances to form a homogeneous molecular dispersion. Quantitatively, it is the concentration of solute in a saturated solution at a specific temperature.



Primary Variables: Physical/chemical properties, Temperature, Pressure, pH, and subdivision state.

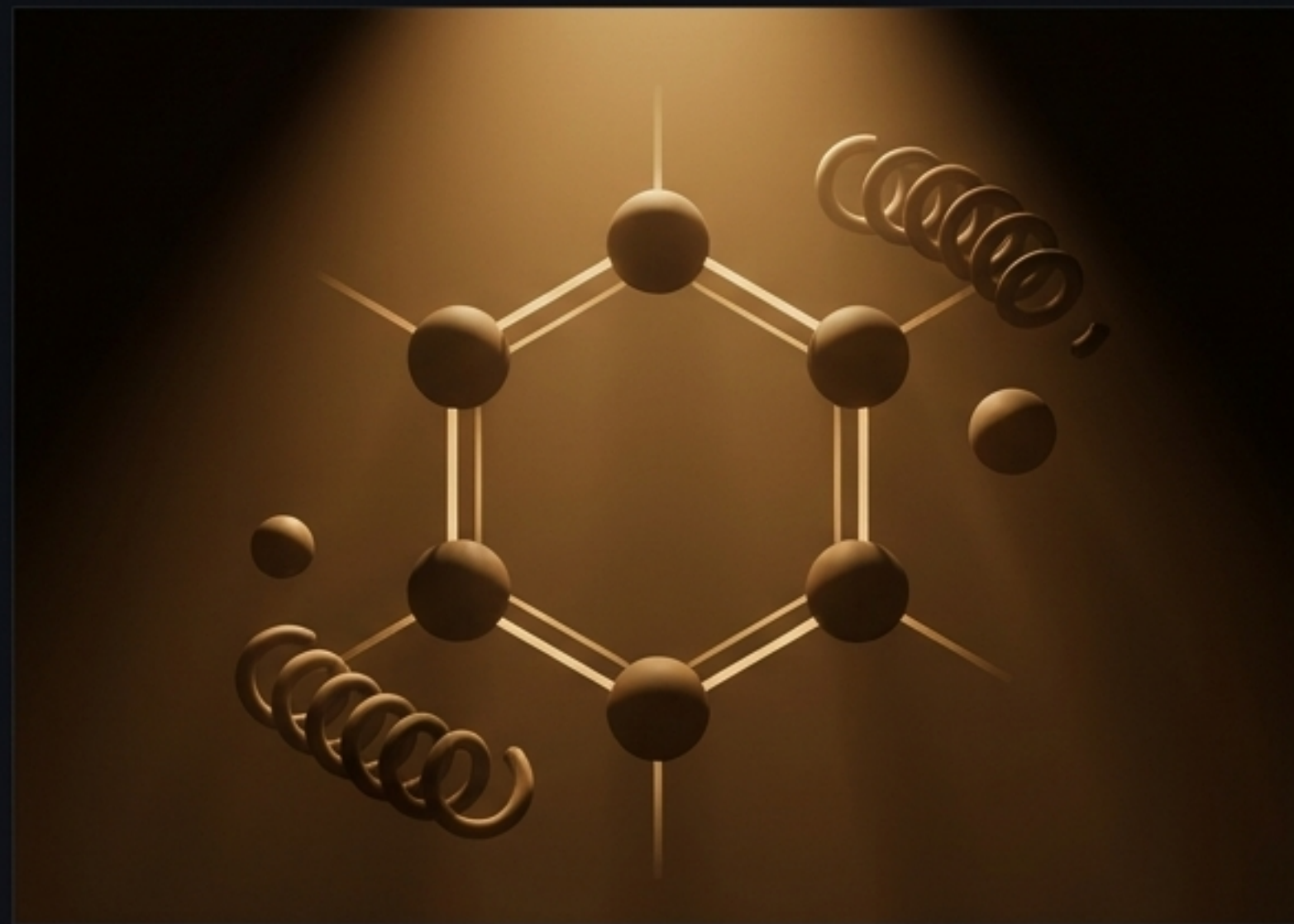
The Universal Rule: “Like Dissolves Like”

Decades of empiric findings summarize solvent-solute interactions into a single fundamental principle.



POLAR REGIME

Water acts as the primary solvent, seamlessly dissolving salts, sugars, and similar polar compounds.



NONPOLAR REGIME

Mineral oil and benzene act as solvents for substances that are practically insoluble in water.

The Solvent Classification Matrix

POLAR SOLVENTS

- High dielectric constant reduces attraction between oppositely charged ions (e.g., NaCl).
- Forms hydrogen bonds to dissolve phenols, alcohols, and aldehydes.
- Breaks covalent bonds of strong electrolytes via acid-base reactions (e.g., HCl in water).

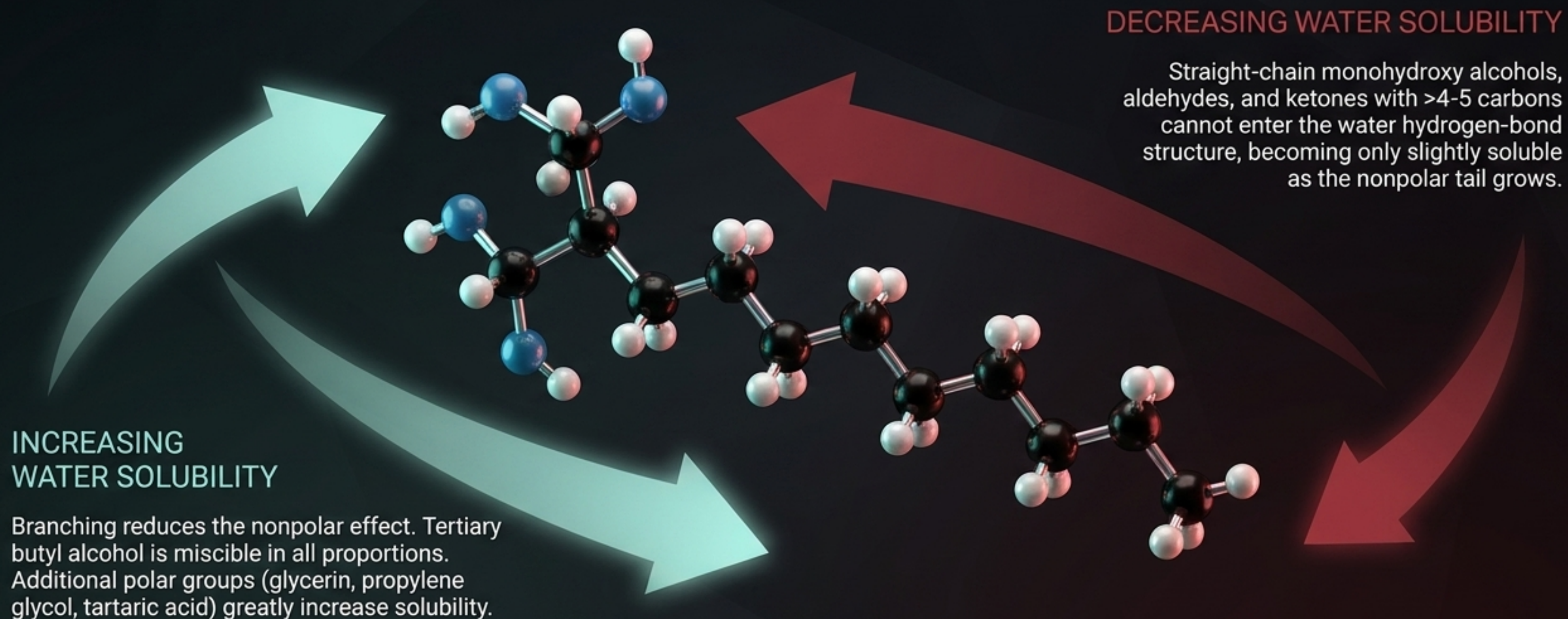
NONPOLAR SOLVENTS

- Low dielectric constant cannot form H-bridges or reduce ion attraction.
- Relies entirely on weak van der Waals-London forces.
- Dissolves oils, fats, and alkaloidal bases.

SEMIPOLAR SOLVENTS

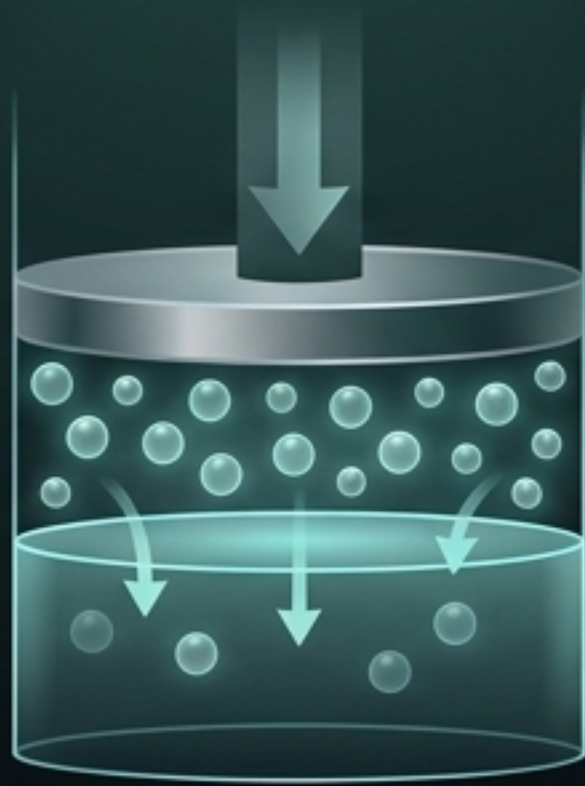
- Ketones and alcohols that induce a degree of polarity in nonpolar molecules.
- Acts as a critical intermediate cosolvent.
- Example: Acetone increases ether solubility in water; Alcohol brings castor oil into water.

The Structural Mechanics of Polarity



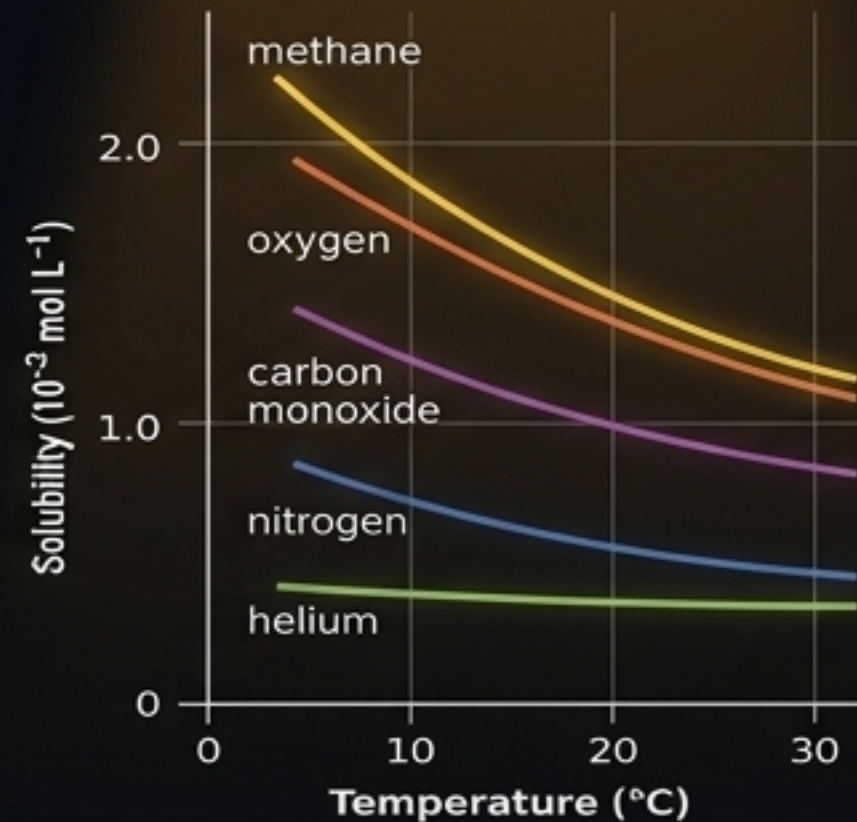
State Dynamics I: Gases in Liquids

Henry's Law (Pressure)



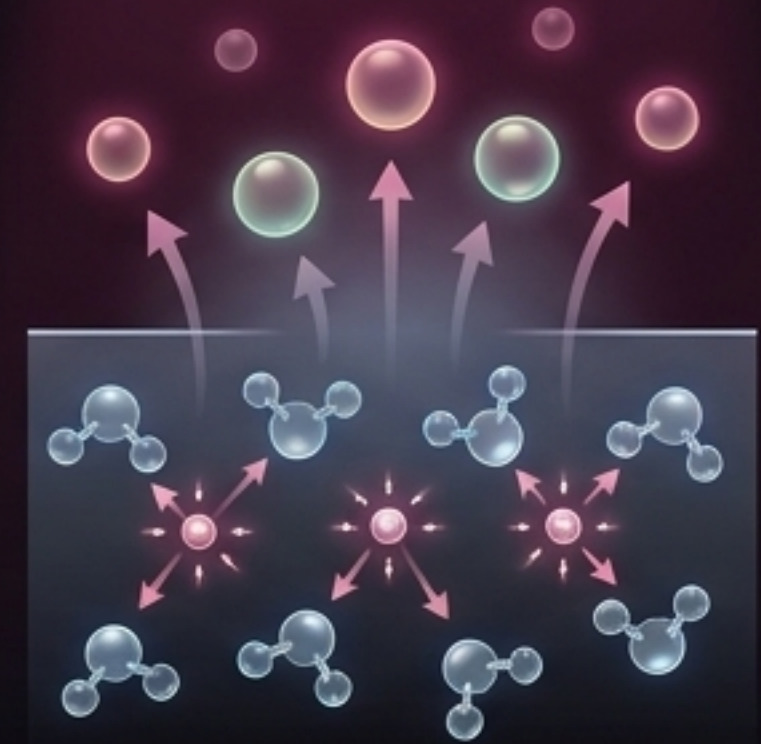
Solubility increases directly as pressure increases at a constant temperature.

Temperature Effect



Increasing temperature decreases solubility as gases tend to escape.

Salting Out

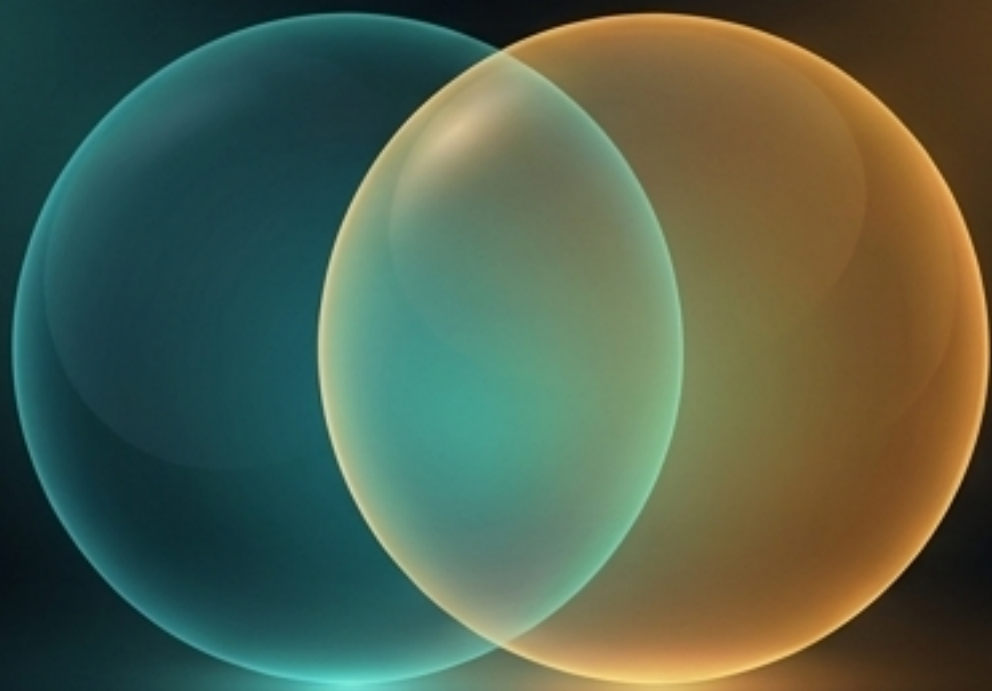


Electrolytes attract water molecules, reducing the density of the aqueous environment adjacent to gas, forcing gas liberation.

Pharmaceutical Applications: Hydrochloric acid, Ammonia water, Effervescent preparations, Aerosol products.

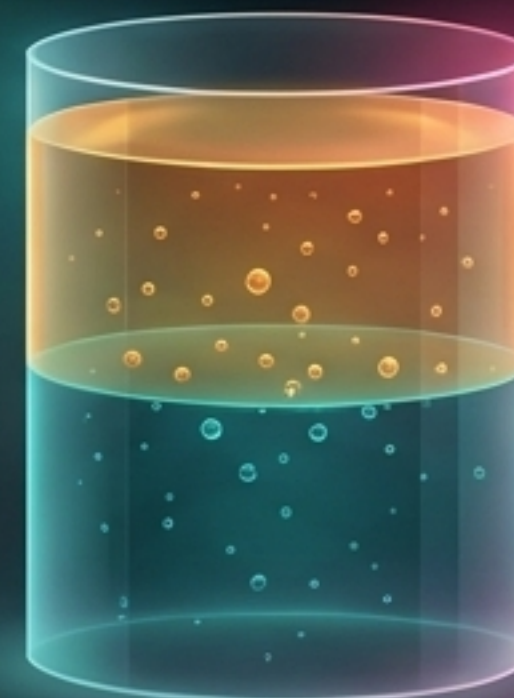
State Dynamics II: Liquid in Liquid Miscibility

Complete Miscibility



Mix in all proportions. Applies to nonpolar solvents (benzene/ CCl_4) and polar/semipolar pairs (water/alcohol).

Partial Miscibility



Forms two distinct liquid layers, each containing a dissolved fraction of the other (e.g., water/ether, water/phenol)

Temperature Dependencies

Miscibility varies by Critical Solution Temperatures (Consolute).

Upper: Phenol-water.

Lower: Soluble below a threshold.

Both: Nicotine-water.

Neither: Ethyl ether-water.

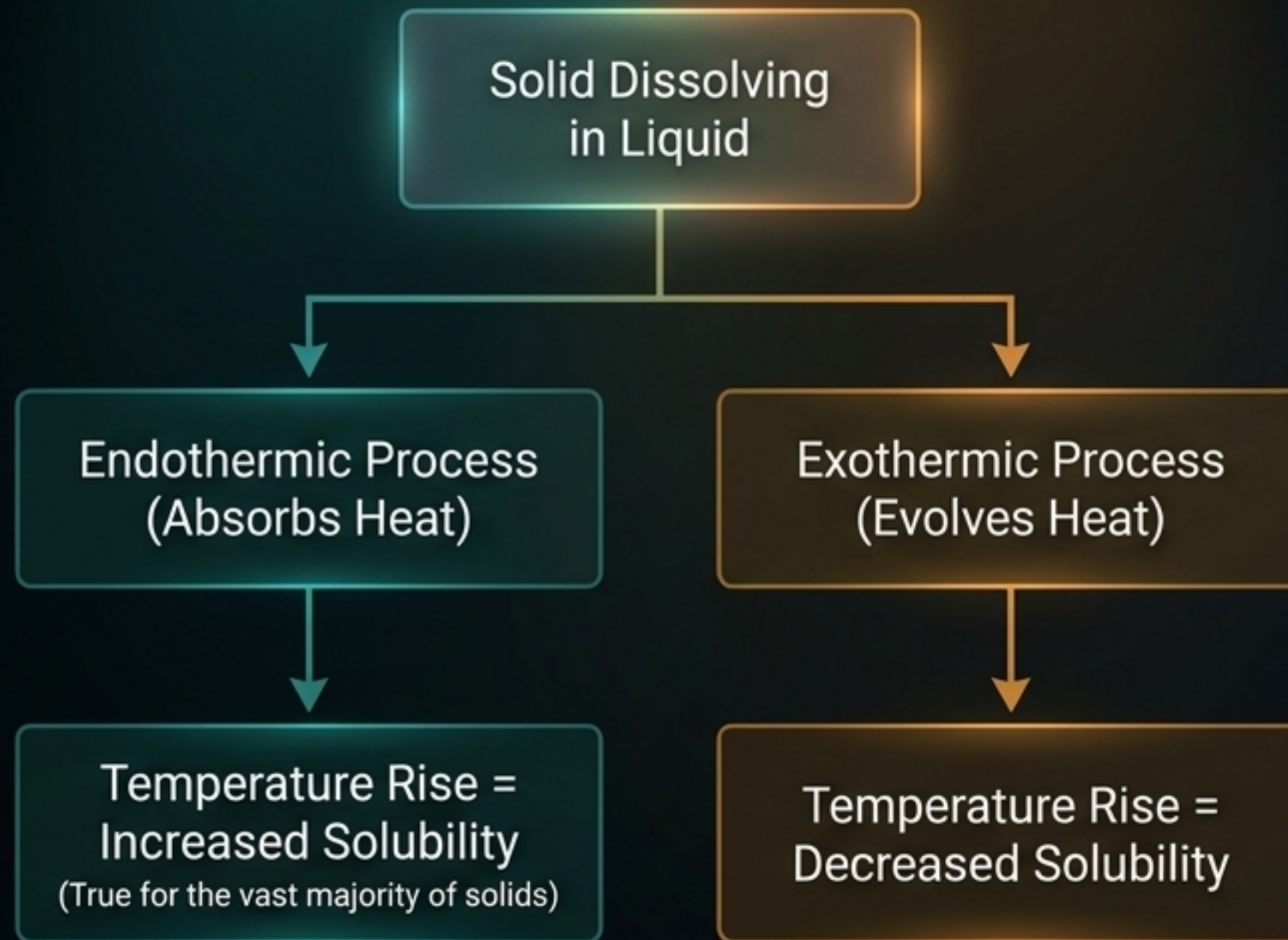
Foreign Substances (Ternary Systems)

Soluble in ONE component = decreases mutual solubility (e.g., naphthalene in phenol/water).

Soluble in BOTH components = increases mutual solubility (e.g., succinic acid in phenol/water).

State Dynamics III: Solids in Liquids

Thermodynamic Temperature Effects



Cosolvency & Complex Formation

- **Cosolvency:** A solute is often vastly more soluble in a strategically blended mixture of solvents (cosolvents) than in one solvent alone.
- **Complex Formation:** The apparent solubility of a solute may be drastically altered (increased or decreased) by adding a third substance that forms intermolecular complexes.

The Nernst Distribution Law



The Partition Coefficient

This structural equilibrium dictates the Nernst Distribution Law, introducing the defining mathematical metric of drug formulation. It explains exactly how molecular preferences scale up to dictate measurable distribution between immiscible liquids.

The Pharmacological Payoff

